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Editors
Constraints

Dear Editors,

Please consider the manuscript *"Salem–Spencer sets as a constraint-solving benchmark: decomposition, certification, and split-policy effects at* $r_3(212)$*"* for publication in *Constraints*.

The manuscript studies a natural finite-domain feasibility family at the frontier of OEIS A003002. It does not claim to resolve $r_3(212)$. Instead, it develops and evaluates a reproducible constraint-solving methodology that combines implied window-cardinality constraints, deterministic cube decomposition, controlled solver comparisons, and independently checkable infeasibility certificates.

The main contributions for the constraint-programming community are:

- A witness-informed CP-SAT architecture in which valid window-cardinality constraints reduce the controlled unresolved rate by about 28 percentage points, substantially more than the generic search interventions tested.
- A matched five-policy split ablation and independent exhaustive cover audit that quantify a cover-size versus conquer-cost tradeoff. Global AP-degree reduces the complete depth-24 cover from 12,582,912 to 96,847 cubes, but produces harder survivors under the historical CP-SAT cap.
- Controlled configuration-level comparisons on released formulas. Native CaDiCaL 3.0.0 and kissat 4.0.4 close all 20 hard-pocket instances on matched hardware; a two-stage CDCL survey closes 6,045 of 6,071 broad residuals and leaves a compact 26-instance challenge tier.
- A certificate and regression pipeline. Selected closures yield LRAT certificates accepted by the formally verified `cake_lpr` checker, while end-to-end tests recover the known exact values $r_3(80) = 22$, $r_3(90) = 24$, and $r_3(100) = 27$ from independently checked lower and upper evidence.

The released benchmark supports research on decomposition, problem-specific implied constraints, search configuration, empirical hardness, and auditable constraint solving. It includes deterministic generators, tiered JSONL and CNF instances, matched solver outputs, model and cover audits, proof objects, hashes, and machine-readable provenance.

Source code and compact artifacts are available in the tagged GitHub snapshot. The versioned dataset is archived on Zenodo, and the preprint is available as arXiv:2606.04016. `ARTIFACT_REPRODUCIBILITY.md` maps the principal claims to their data and reproduction commands; the $N = 80$ exact-value regression is the fastest complete check and runs in minutes on a workstation. The manuscript is not under consideration elsewhere.

Generative-AI use is disclosed in the manuscript. The author made all final scientific and experimental decisions, reviewed the generated text and code, independently checked the reported claims, and takes full responsibility for the work. The author declares no competing interests.

Thank you for your consideration.

Sincerely,

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